

STATISTICS REVIEW SUMMARY

Regression, Correlation, and Hypothesis Testing Overview

CHAPTER OVERVIEW

Unit 1: Correlation & Linear Relationships

- Measures relationship strength and direction (r)
- Visual tool: scatterplot (positive, negative, none)
- r -values: ± 1 = perfect line, 0 = no linear trend

Unit 2: Regression (Least Squares Line)

- Regression line equation: $\hat{y} = a + bx$
- Slope (b): change in y for 1-unit x increase
- Intercept (a): predicted y when $x = 0$
- Residuals = observed - predicted

Unit 3: Regression Assumptions

1. Linearity: relationship follows straight line
2. Independence: observations are unrelated
3. Constant Variability: even spread of residuals
4. Nearly Normal Residuals: residuals roughly symmetric

Unit 4: Outliers, Leverage & Influence

- Leverage: extreme x -values (far from mean of x)
- Residual: distance between observed y and predicted y
- Influential Point: both high leverage and large residual

Unit 5: Hypothesis Testing for Regression

- Tests whether slope (b_1) differs from 0
- $t = b / SE(b)$
- Large $|t|$ $\rightarrow$ small p $\rightarrow$ reject H_0
- H_0 : $b_1 = 0$ (no relationship)
- H_a : $b_1 \neq 0$ (relationship exists)

KEY FORMULAS

$$r = (1/(n-1)) * \sum((x_i - \bar{x})(y_i - \bar{y}) / (s_x * s_y)) \quad \text{Correlation}$$

$b = r * (s_y/s_x)$ Slope
 $a = \bar{y} - b*\bar{x}$ Intercept
 $R^2 = r^2$ Coefficient of Determination
 $R^2 \text{ adj} = 1 - (1 - R^2)*((n-1)/(n-k-1))$ Adjusted R^2
 $t = b / SE(b)$ t-Statistic for slope
 $p = P(|T| > |t|)$ p-value
 $e = y - \hat{y}$ Residual

KEY CONCEPTS

Slope (b_1): change in y for each 1-unit x
Intercept (b_0): predicted y when $x=0$
Residual: observed - predicted y
Leverage: distance of x from mean x
Influence: effect of point on slope
t-value: number of SEs slope differs from 0
p-value: probability slope = 0
 R^2 : percent of y variation explained by x

DIFFICULT CONCEPTS

1. Correlation does not imply causation
2. High Residual does not mean High Leverage
3. Interpreting t-values ($t = b / SE(b)$)
4. R^2 vs Adjusted R^2
5. Extrapolation beyond data range

HYPOTHESIS TEST SUMMARY

1. Write hypotheses: $H_0: b_1=0$, $H_a: b_1 \neq 0$
2. Compute $t = b / SE(b)$
3. Find p-value
4. Decision: $p < 0.05 \rightarrow$ reject H_0
5. Interpretation: Variable predicts outcome

FINAL TAKEAWAYS

- Correlation (r) shows linear strength and direction

- Regression predicts y from x
- Slope = rate of change; Intercept = baseline
- t-test checks if slope $\neq 0$
- Small p-value = strong evidence of relationship
- R^2 shows explanatory power
- Avoid extrapolation; check assumptions
- High leverage points can change slope drastically